Math 242, S20
Exam 2

Name: Answer

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	20	
2	20	
3	24	
4	18	
5	18	
Total	100	

Good Luck !

Problem 1. (20 points) Given the following homogeneous second-order differential equation:

$$y'' + 4y' + 13y = 0.$$

(a) Find the general solution of the given differential equation.

Solution. Characteristic equation

$$r^2+4r+13=0$$

$$r=\frac{-4\pm\sqrt{4^2-4\cdot1\cdot13}}{2}$$

$$=\frac{-4\pm\sqrt{-36}}{2}=-2\pm3i$$

$$\Rightarrow y(x)=e^{-2x}(C_1\cos3x+C_2\sin3x)$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 3, \quad y'(0) = 0.$$

Solution.

$$y'(x)=(-2)e^{-2x}(C_1\cos3x+C_2\sin3x)$$
$$+e^{-2x}(-3C_1\sin3x+3C_2\cos3x)$$

$$=e^{-2x}[C_1(-2\cos3x-3\sin3x)$$
$$+C_2(-2\sin3x+3\cos3x)$$

$$y(0)=3 \Rightarrow C_1=3$$

$$y'(0)=0 \Rightarrow -2C_1+3C_2=0 \Rightarrow C_2=\frac{2}{3}C_1=2$$

Thus $y(x)=e^{-2x}(3\cos3x+2\sin3x)$

Problem 2. (20 points) Given a nonhomogeneous linear third-order differential equation:

$$y^{(3)} - 6y'' + 9y' = 9x^2 - 15$$

(a) Find the complementary solution y_c of the above nonhomogeneous equation (i.e., the general solution of the associated homogeneous equation).

Solution: Characteristic equation

$$r^3 - 6r^2 + 9r = 0$$

$$r(r^2 - 6r + 9) = 0$$

$$r(r-3)^2 = 0$$

$$\Rightarrow r_1 = 0, \quad r_2 = r_3 = 3$$

$$\Rightarrow y_c(x) = C_1 + (C_2 + C_3 x)e^{3x}$$

(b) Then find a particular solution y_p of the nonhomogeneous equation by the method of undetermined coefficient.

Solution: Since $f(x) = 9x^2 - 15$, we have

$$y_p(x) = x^s(Ax^2 + Bx + C)$$

$\rightarrow s = 1$ to avoid duplications of terms in $y_c(x)$

$$= x(Ax^2 + Bx + C) = Ax^3 + Bx^2 + Cx$$

$$\left.\begin{aligned} y_p'(x) &= 3Ax^2 + 2Bx + C \\ y_p''(x) &= 6Ax + 2B \\ y_p^{(3)}(x) &= 6A \end{aligned}\right\} \Rightarrow \begin{aligned} &y_p^{(3)} - 6y_p'' + 9y_p' \\ &= 6A - 6(6Ax + 2B) \\ &\quad + 9(3Ax^2 + 2Bx + C) \\ &= 27Ax^2 + (-36A + 18B)x \\ &\quad + (6A - 12B + 9C) \equiv 9x^2 - 15 \end{aligned}$$

$$\Rightarrow \begin{cases} 27A = 9 \Rightarrow A = \frac{1}{3} \\ -36A + 18B = 0 \Rightarrow B = 2A = \frac{2}{3} \\ 6A - 12B + 9C = -15 \Rightarrow 2 - 8 + 9C = -15 \Rightarrow C = -1 \end{cases}$$

thus $y_p(x) = \frac{1}{3}x^3 + \frac{2}{3}x^2 - x$

Problem 3. (24 points)

(a) $\mathcal{L}\{\cosh^2 2t\}$. (Hint: *use* $\cosh 2t = \frac{e^{2t}-e^{-2t}}{2}$ *to compute* $\cosh^2 2t$)

$$\cosh^2 2t = \left(\frac{e^{2t}+e^{-2t}}{2}\right)^2 = \frac{1}{4}(e^{4t}+2+e^{-4t})$$

$$\Rightarrow \mathcal{L}\{\sinh^2 2t\} = \mathcal{L}\{\tfrac{1}{4}(e^{4t}+e^{-4t}+2\}$$

$$= \tfrac{1}{4}(\mathcal{L}\{e^{4t}\}+\mathcal{L}\{e^{-4t}\}+\mathcal{L}\{2\})$$

$$= \tfrac{1}{4}\left(\frac{1}{s-4}+\frac{1}{s+4}+\frac{2}{s}\right), \quad s>4.$$

$$= \frac{s^2-8}{s(s^2-16)}$$

(b) Find $\mathcal{L}^{-1}\left\{\frac{1}{s(s-3)^2}\right\}$ by using the formula for **Transform of Integrals**.

$$F(s) = \frac{1}{(s-3)^2} \Rightarrow f(t) = \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-3)^2}\right\} = te^{3t}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s-3)^2}\right\} = \int_0^t f(\tau)\,d\tau = \int_0^t \tau e^{3\tau}d\tau$$

$$= \int_0^t \tau\left(\frac{e^{3\tau}}{3}\right)' d\tau = \left[\frac{\tau e^{3\tau}}{3}\right]_{\tau=0}^{\tau=t} - \int_0^t \frac{e^{3\tau}}{3}d\tau$$

$$= \frac{te^{3t}}{3} - \left[\frac{e^{3\tau}}{9}\right]_{\tau=0}^{\tau=t} = \frac{te^{3t}}{3} - \frac{e^{3t}}{9} + \frac{1}{9}$$

(c) Find $\mathcal{L}^{-1}\left\{\frac{s+8}{s^2+16}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{s+8}{s^2+16}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} + \mathcal{L}^{-1}\left\{\frac{8}{s^2+16}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{s}{s^2+4^2}\right\} + 2\mathcal{L}^{-1}\left\{\frac{4}{s^2+4^2}\right\}$$

$$= \cos 4t + 2\sin 4t$$

Problem 4. (18 points) Use **Laplace transform** to solve the initial value problem:

$$x'' - 16x = 0; \quad x(0) = 3,\ x'(0) = 6.$$

Solution Let $X(s) = \mathcal{L}\{x(t)\}$, then

$$\mathcal{L}\{x''(t)\} = s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0)$$
$$= s^2 X(s) - 3s - 6$$

Thus

$$\mathcal{L}\{x'' - 16x\} = \mathcal{L}\{0\}$$
$$\mathcal{L}\{x''\} - 16\mathcal{L}\{x\} = 0$$
$$(s^2 X(s) - 3s - 6) - (16 X(s)) = 0$$
$$(s^2 - 16) X(s) = 3s + 6$$
$$X(s) = \frac{3s+6}{s^2-16}$$

$$\frac{3s+6}{s^2-16} = \frac{3s+6}{(s+4)(s-4)} \equiv \frac{A}{s+4} + \frac{B}{s-4}$$
$$= \frac{A(s-4)+B(s+4)}{(s+4)(s-4)} = \frac{(A+B)s+(-4A+4B)}{(s+4)(s-4)}$$

which gives us

$$\begin{cases} A+B=3 \\ -4A+4B=6 \end{cases} \Rightarrow \begin{matrix} A = \frac{3}{4} \\ B = \frac{9}{4} \end{matrix}$$

Thus $x(t) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{3}{4}\frac{1}{s+4} + \frac{9}{4}\frac{1}{s-4}\right\}$

$$= \frac{3}{4}\mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} + \frac{9}{4}\mathcal{L}^{-1}\left\{\frac{1}{s-4}\right\}$$
$$= \frac{3}{4}e^{-4t} + \frac{9}{4}e^{4t}$$

(or $x(t) = 3\cosh 4t + \frac{3}{2}\sinh 4t$)

Problem 5. (18 points) Use **Laplace transform** to solve the initial value problem:

$$x'' + 4x' - 5x = 2; \quad x(0) = x'(0) = 0.$$

Solution: let $X(s) = \mathcal{L}\{x(t)\}$, then

$$\mathcal{L}\{x'(t)\} = sX(s) + x(0) = sX(s)$$

$$\mathcal{L}\{x''(t)\} = s^2 X(s) - s\,x(0) - x'(0) = s^2 X(s)$$

Then

$$\mathcal{L}\{x'' + 4x' - 5x\} = \mathcal{L}\{2\}$$

$$\mathcal{L}\{x''\} + 4\mathcal{L}\{x'\} - 5\mathcal{L}\{x\} = \frac{2}{s}$$

$$s^2 X(s) + 4s X(s) - 5X(s) = \frac{2}{s}$$

$$(s^2 + 4s - 5) X(s) = \frac{2}{s}$$

$$\Rightarrow X(s) = \frac{2}{s(s^2+4s-5)}$$

$$\frac{2}{s(s^2+4s-5)} = \frac{2}{s(s-1)(s+5)} \equiv \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+5}$$

$$= \frac{A(s-1)(s+5) + Bs(s+5) + Cs(s-1)}{s(s-1)(s+5)}$$

let $s=0$, we get $2 = A(0-1)(0+5) \Rightarrow A = -\frac{2}{5}$

let $s=1$, we get $2 = B \cdot 1 \cdot (1+5) \Rightarrow B = \frac{1}{3}$

let $s=-5$, we get $2 = C \cdot (-5)(-5-1) \Rightarrow C = \frac{1}{15}$

Thus

$$x(t) = \mathcal{L}^{-1}\left\{ -\frac{2}{5}\frac{1}{s} + \frac{1}{3}\frac{1}{s-1} + \frac{1}{15}\frac{1}{s+5} \right\}$$

$$= -\frac{2}{5} + \frac{1}{3}e^{t} + \frac{1}{15}e^{-5t}$$